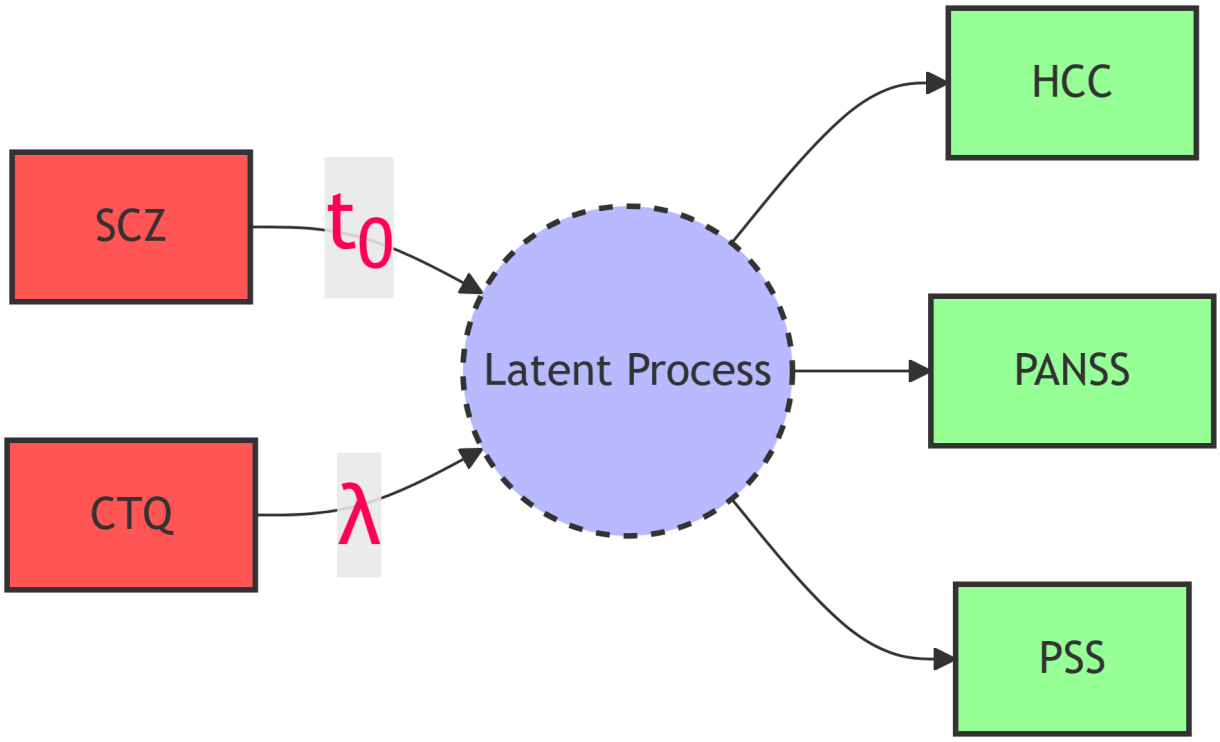
# Supplementary material

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### Causal structure




### Model formulation

#### 1. Latent childhood trauma metrics

For each subject $i$, the joint distribution of their underlying exposure to abuse ($c_{i,1}$) and neglect ($c_{i,2}$) is drawn from a centered multivariate normal distribution capturing population variance and structural trait correlation:

$$\begin{bmatrix} c_{i,1} \\ c_{i,2} \end{bmatrix} \sim \text{Multivariate-Normal} \left( \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \boldsymbol{\Sigma}_c \right)$$

$$\boldsymbol{\Sigma}_c = \text{diag } (\boldsymbol{\sigma}_c) \, \boldsymbol{\Omega}_c \, \text{diag } (\boldsymbol{\sigma}_c)$$

Where $\boldsymbol{\sigma}_c = [\sigma_{\text{abuse}}, \sigma_{\text{neglect}}]^T$ maps the population scales, and $\boldsymbol{\Omega}_c$ represents the trait correlation matrix.

**2. CTQ measurement (emission) layer**

The individual's true latent trauma states directly manifest across the five observed subscales of the Childhood Trauma Questionnaire. The Abuse factor ($c_{i,1}$) drives emotional, physical, and sexual dimensions, while the Neglect factor ($c_{i,2}$) drives emotional and physical neglect indicators:

$$
\begin{aligned}
y_m^{(\mathrm{EA})} &\sim \mathrm{Normal}\left(\alpha_{\mathrm{EA}} + c_{\mathrm{subj}\,[m],1}, \sigma_{\mathrm{EA}}\right) \quad \text{[Emotional Abuse]} \\
y_m^{(\mathrm{PA})} &\sim \mathrm{Normal}\left(\alpha_{\mathrm{PA}} + c_{\mathrm{subj}\,[m],1}, \sigma_{\mathrm{PA}}\right) \quad \text{[Physical Abuse]} \\
y_m^{(\mathrm{SA})} &\sim \mathrm{Normal}\left(\alpha_{\mathrm{SA}} + c_{\mathrm{subj}\,[m],1}, \sigma_{\mathrm{SA}}\right) \quad \text{[Sexual Abuse]} \\
y_m^{(\mathrm{EN})} &\sim \mathrm{Normal}\left(\alpha_{\mathrm{EN}} + c_{\mathrm{subj}\,[m],2}, \sigma_{\mathrm{EN}}\right) \quad \text{[Emotional Neglect]} \\
y_m^{(\mathrm{PN})} &\sim \mathrm{Normal}\left(\alpha_{\mathrm{PN}} + c_{\mathrm{subj}\,[m],2}, \sigma_{\mathrm{PN}}\right) \quad \text{[Physical Neglect]}
\end{aligned}
$$

**3. Latent dynamics**

The latent dysregulation state $z_i(t)$ transitions over irregular time intervals ($\Delta t = t - t'$) according to a continuous-time (Ornstein–Uhlenbeck) engine. The individual recovery speed $\lambda_i$ forces a strict positive bound via the structural baseline floor $\lambda_0$:

$$
\begin{aligned}
z_i(t) \mid z_i(t') &\sim \mathrm{Normal}\left(e^{-\lambda_i \Delta t}\, z_i(t'), \sigma_z \sqrt{1 - e^{-2\lambda_i \Delta t}}\right) \\
\lambda_i &= \lambda_0 + \exp\left(\gamma_{\mathrm{abuse}}\, c_{i,1} + \gamma_{\mathrm{neglect}}\, c_{i,2}\right)
\end{aligned}
$$

**4. Longitudinal process state**

The state trajectory begins at a unique initial tracking baseline anchored to a matched-pair intercept ($u_{\mathrm{pair}}$) and shifted by diagnostic status (SCZ):

$$
\begin{aligned}
z_i(t_0) &\sim \mathrm{Normal}\left(u_{\mathrm{pair}\,[i]} + \beta_{\mathrm{SCZ}}\, \mathrm{SCZ}_i, 1\right) \\
u_j &\sim \mathrm{Normal}\left(0, \sigma_{\mathrm{pair}}\right)
\end{aligned}
$$

The evolving state $z_i(t)$ projects concurrent signals down into both historical biological archives and acute clinical cross-sections:

$$
\begin{aligned}
y_m^{(\mathrm{HCC})} &\sim \text{Student-t}\left(\nu = 6, a_{\mathrm{HCC}} + b_{\mathrm{HCC}}\, z_{\mathrm{state}\,[m]}, \sigma_{\mathrm{HCC}}\right) \\
y_m^{(\mathrm{PANSS})} &\sim \mathrm{Normal}\left(a_{\mathrm{PANSS}} + b_{\mathrm{PANSS}}\, z_{\mathrm{state}\,[m]}, \sigma_{\mathrm{PANSS}}\right) \\
y_m^{(\mathrm{PSS})} &\sim \mathrm{Normal}\left(a_{\mathrm{PSS}} + b_{\mathrm{PSS}}\, z_{\mathrm{state}\,[m]}, \sigma_{\mathrm{PSS}}\right)
\end{aligned}
$$

**5. Prior distributions**

$$
\begin{aligned}
\lambda_0, \gamma_{\mathrm{abuse}}, \gamma_{\mathrm{neglect}}, \beta_{\mathrm{SCZ}} &\sim \mathrm{Normal}\ (0, 1) \\
\alpha_{\mathrm{EA}}, \alpha_{\mathrm{PA}}, \alpha_{\mathrm{SA}}, \alpha_{\mathrm{EN}}, \alpha_{\mathrm{PN}} &\sim \mathrm{Normal}\ (0, 1) \\
a_{\mathrm{PANSS}}, a_{\mathrm{PSS}}, a_{\mathrm{HCC}} &\sim \mathrm{Normal}\ (0, 0.5) \\
b_{\mathrm{PANSS}}, b_{\mathrm{PSS}}, b_{\mathrm{HCC}} &\sim \mathrm{Normal}\ (0, 1) \\
\sigma_z, \sigma_{\mathrm{pair}}, \sigma_{\mathrm{HCC}}, \sigma_{\mathrm{PANSS}}, \sigma_{\mathrm{PSS}} &\sim \mathrm{Exponential}\ (1) \\
\sigma_{\mathrm{EA}}, \sigma_{\mathrm{PA}}, \sigma_{\mathrm{SA}}, \sigma_{\mathrm{EN}}, \sigma_{\mathrm{PN}} &\sim \mathrm{Exponential}\ (1) \\
\boldsymbol{\sigma}_c &\sim \mathrm{Exponential}\ (1) \\
\boldsymbol{\Omega}_c &\sim \mathrm{LKJ}\ (\eta = 2)
\end{aligned}
$$

**Stan code**

The mathematical model is rewritten in the Stan language.

```
data{

int <lower=1> N_state;
```

```
array[N_state] int <lower=1> subj_id;
int <lower=1> N_pair;
array[N_state] int <lower=1> pair_id;
vector[N_state] time;
array[N_state] int <lower=0, upper=1> is_first;
array[N_state] int <lower=0, upper=1> scz;

int <lower=1> N_hcc;
array[N_hcc] int <lower=1> state_hcc;
vector[N_hcc] y_hcc;

int <lower=1> N_panss;
array[N_panss] int <lower=1> state_panss;
vector[N_panss] y_panss;

int <lower=1> N_pss;
array[N_pss] int <lower = 1> state_pss;
vector[N_pss] y_pss;

int <lower=1> N_ctq;
int <lower=1> N_subj;
array[N_ctq] int <lower=1> subj_ctq;

vector[N_ctq] y_ea;
vector[N_ctq] y_pa;
vector[N_ctq] y_sa;
vector[N_ctq] y_en;
vector[N_ctq] y_pn;


}

parameters{

vector[N_state] z_raw;
real <lower=0> lambda_0;
real gamma_abuse;
real gamma_neglect;
real <lower=0> sigma_z;
real <lower=0> sigma_pair;

vector[N_pair] u_pair;

real beta_scz;

real a_hcc;
real <lower=0> b_hcc;
real <lower=0> sigma_hcc;

real a_panss;
real b_panss;
real <lower=0> sigma_panss;

real a_pss;
real b_pss;
real <lower=0> sigma_pss;
```

```
real <lower=0> sigma_ea;
real <lower=0> sigma_pa;
real <lower=0> sigma_sa;
real <lower=0> sigma_en;
real <lower=0> sigma_pn;


real alpha_ea;
real alpha_pa;
real alpha_sa;
real alpha_en;
real alpha_pn;

matrix[2, N_subj] c_raw;
vector <lower=0> [2] sigma_c;
cholesky_factor_corr[2] L_c;

}

transformed parameters {

// Covariance matrix for CTQ latent factor c1 (abuse) and c2 (neglect)
matrix[2, N_subj] c;
c = diag_pre_multiply(sigma_c, L_c) * c_raw;

// latent subject-level factors c[1, ] and c[2, ]. Implicit Bayesian imputation.
vector[N_subj] lambda_subj;

for (i in 1:N_subj) {
  lambda_subj[i] = lambda_0 + exp(gamma_abuse * c[1, i] + gamma_neglect * c[2, i]);
}

// dynamic latent process
vector[N_state] z;
for (k in 1:N_state) {
    // first point in time aligned between patients and controls
    if (is_first[k] == 1) {
        z[k] = u_pair[pair_id[k]] + beta_scz * scz[k] + 1 * z_raw[k];
    // Ornstein-Uhlenbeck process (exponential return with stochastic volatility)
    } else {
        real dt = time[k] - time[k-1];
        real phi = exp(-lambda_subj[subj_id[k]] * dt);
        real sigma_dt = sigma_z * sqrt(1 - square(phi));
        z[k] = phi * z[k-1] + sigma_dt * z_raw[k];
    }
  }


}

model{

// priors
lambda_0 ~ normal(0, 1);
gamma_abuse ~ normal(0, 1);
```

```
gamma_neglect ~ normal(0, 1);
sigma_z ~ exponential(1);
sigma_pair ~ exponential(1);
sigma_hcc ~ exponential(1);
sigma_panss ~ exponential(1);
sigma_pss ~ exponential(1);
beta_scz ~ normal(0, 1);
a_hcc ~ normal(0, 0.5);
b_hcc ~ normal(0, 1);
a_panss ~ normal(0, 0.5);
b_panss ~ normal(0, 1);
a_pss ~ normal(0, 0.5);
b_pss ~ normal(0, 1);
z_raw ~ std_normal();

u_pair ~ normal(0, sigma_pair);

sigma_ea ~ exponential(1);
sigma_pa ~ exponential(1);
sigma_sa ~ exponential(1);
sigma_en ~ exponential(1);
sigma_pn ~ exponential(1);

alpha_ea ~ normal(0, 1);
alpha_pa ~ normal(0, 1);
alpha_sa ~ normal(0, 1);
alpha_en ~ normal(0, 1);
alpha_pn ~ normal(0, 1);

to_vector(c_raw) ~ std_normal();

L_c ~ lkj_corr_cholesky(2);


// hcc emissions
for (m in 1:N_hcc) {
    y_hcc[m] ~ student_t(6, a_hcc + b_hcc * z[state_hcc[m]], sigma_hcc);
    }

// panss emissions
for (m in 1:N_panss) {
    y_panss[m] ~ normal(a_panss + b_panss * z[state_panss[m]], sigma_panss);
    }

// pss emissions
for (m in 1:N_pss) {
    y_pss[m] ~ normal(a_pss + b_pss * z[state_pss[m]], sigma_pss);
    }

// CTQ emissions
y_ea ~ normal(alpha_ea + c[1, subj_ctq]', sigma_ea);
y_sa ~ normal(alpha_sa + c[1, subj_ctq]', sigma_sa);
y_pa ~ normal(alpha_pa + c[1, subj_ctq]', sigma_pa);
y_en ~ normal(alpha_en + c[2, subj_ctq]', sigma_en);
y_pn ~ normal(alpha_pn + c[2, subj_ctq]', sigma_pn);
```

```
}

generated quantities{

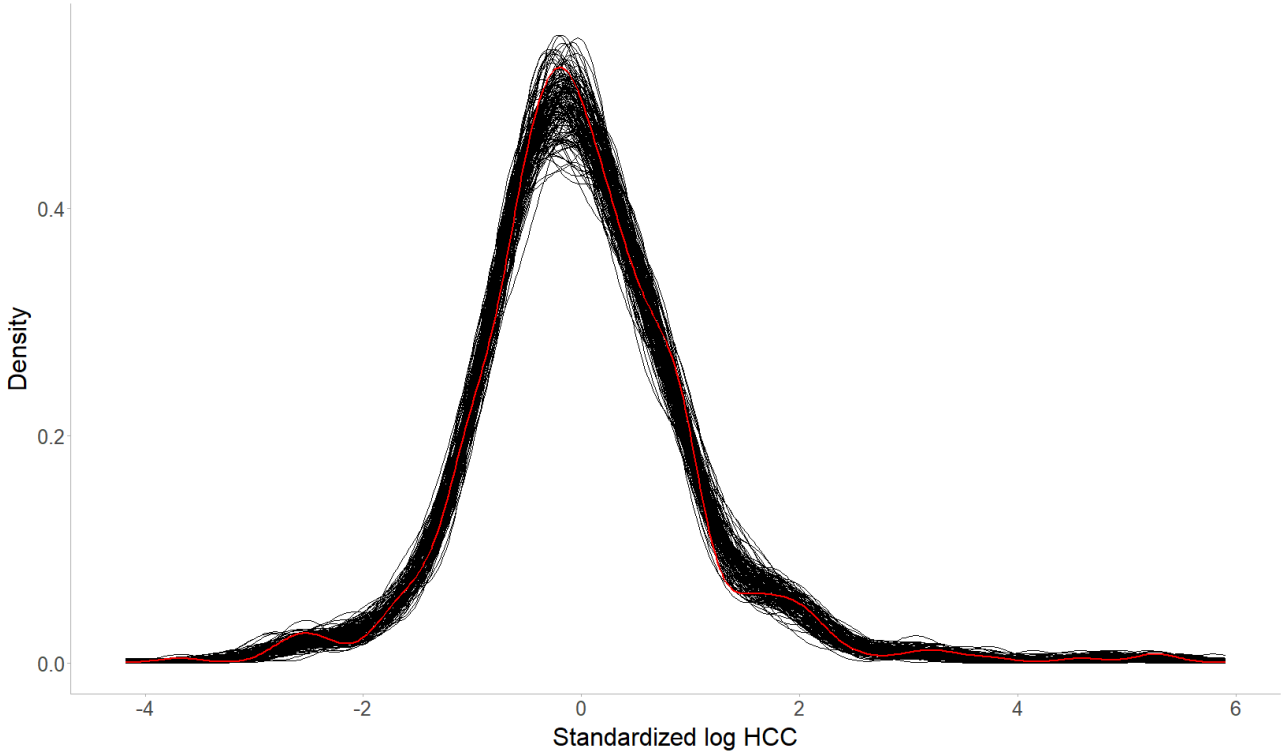

matrix[2, 2] corr = multiply_lower_tri_self_transpose(L_c); // Correlation matrix
matrix[2, 2] cov = quad_form_diag(corr, sigma_c); // (S * R * S) Covariance matrix

}
```

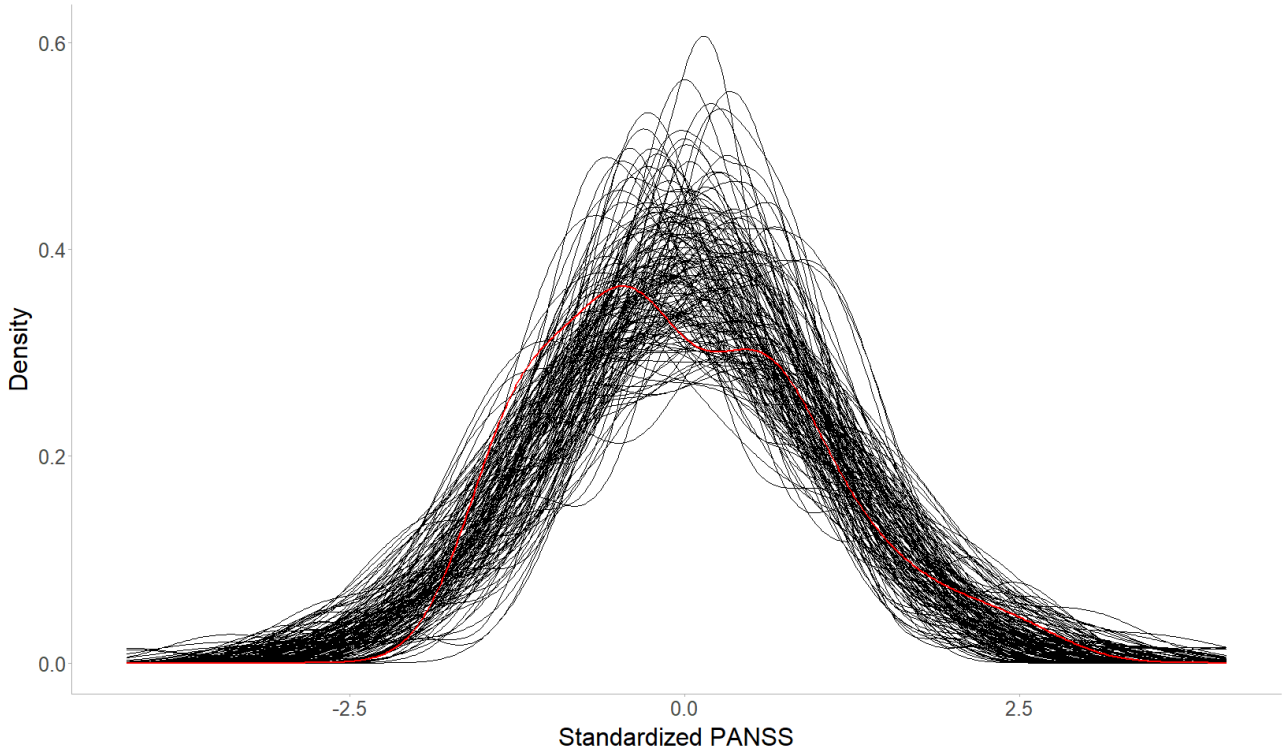
### Model diagnostics and posterior predictive checks

The model was rigorously evaluated for convergence using four Markov chains with 4000 iterations each. The Rhat diagnostic was used to assess convergence, with a value below 1.01 suggesting adequate convergence. The effective sample size (ESS) was over 400 for each parameter. The posterior predictive checks follow.

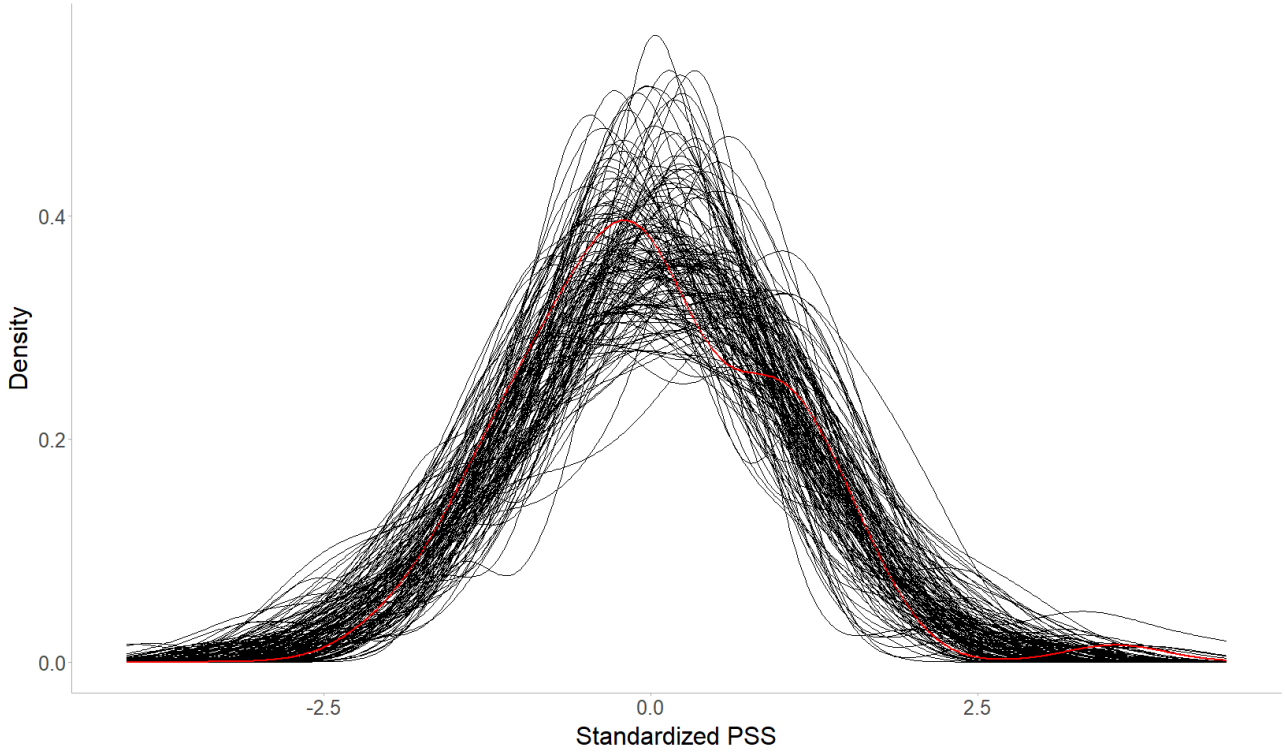
### Density



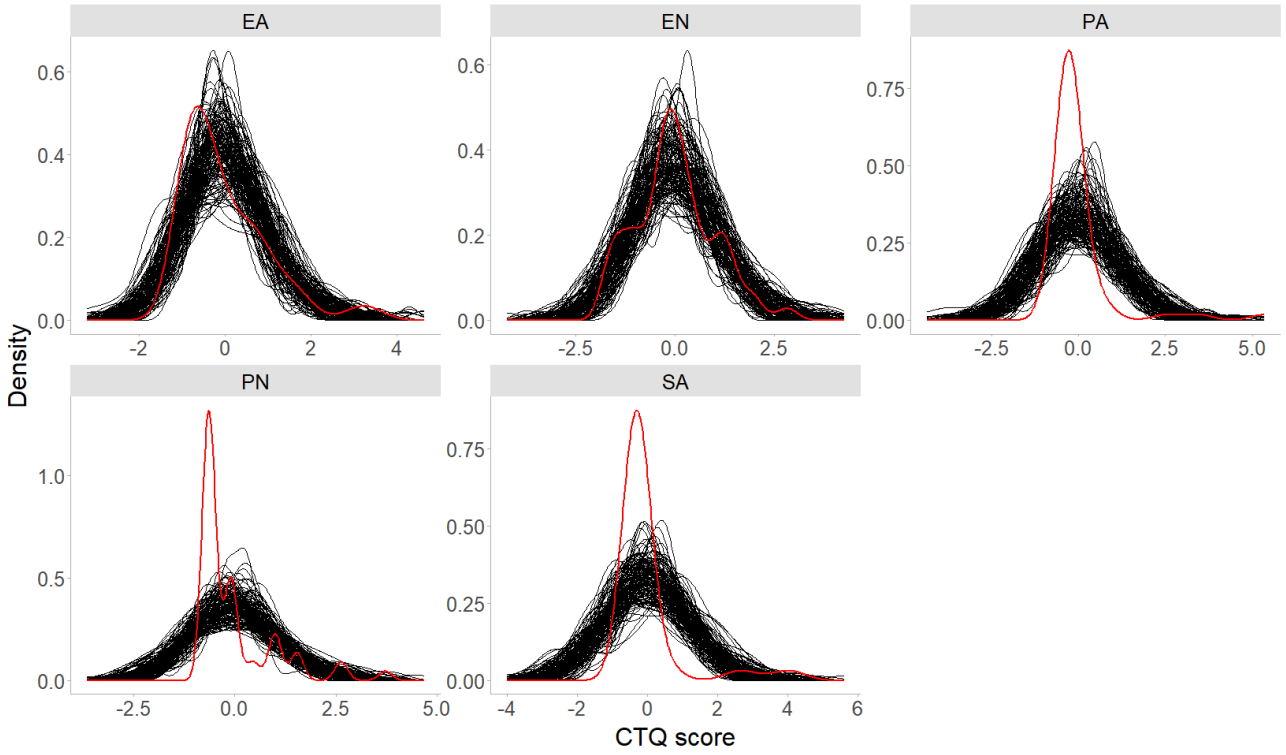

S_Figure 1: Posterior predictive check of HCC. The red line shows the observed values.




S_Figure 2: Posterior predictive check of PANSS. The red line shows the observed values.

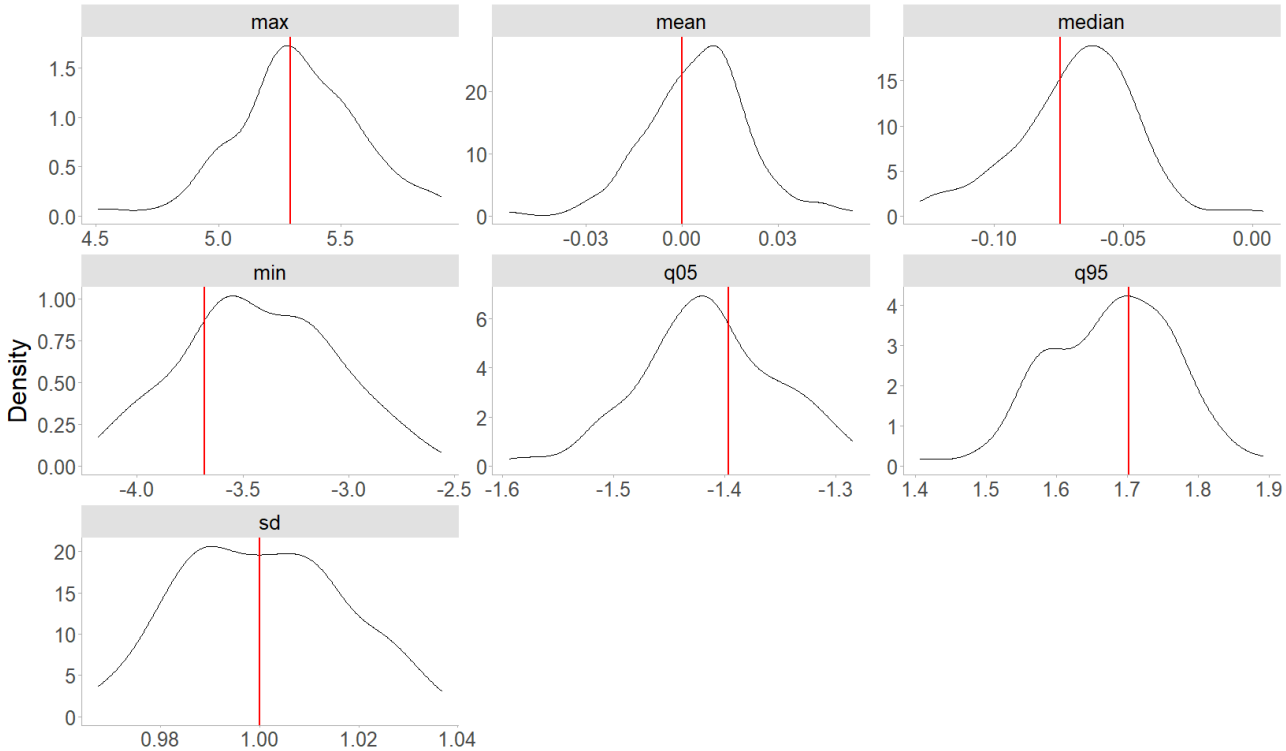



S_Figure 3: Posterior predictive check of PSS. The red line shows the observed values.

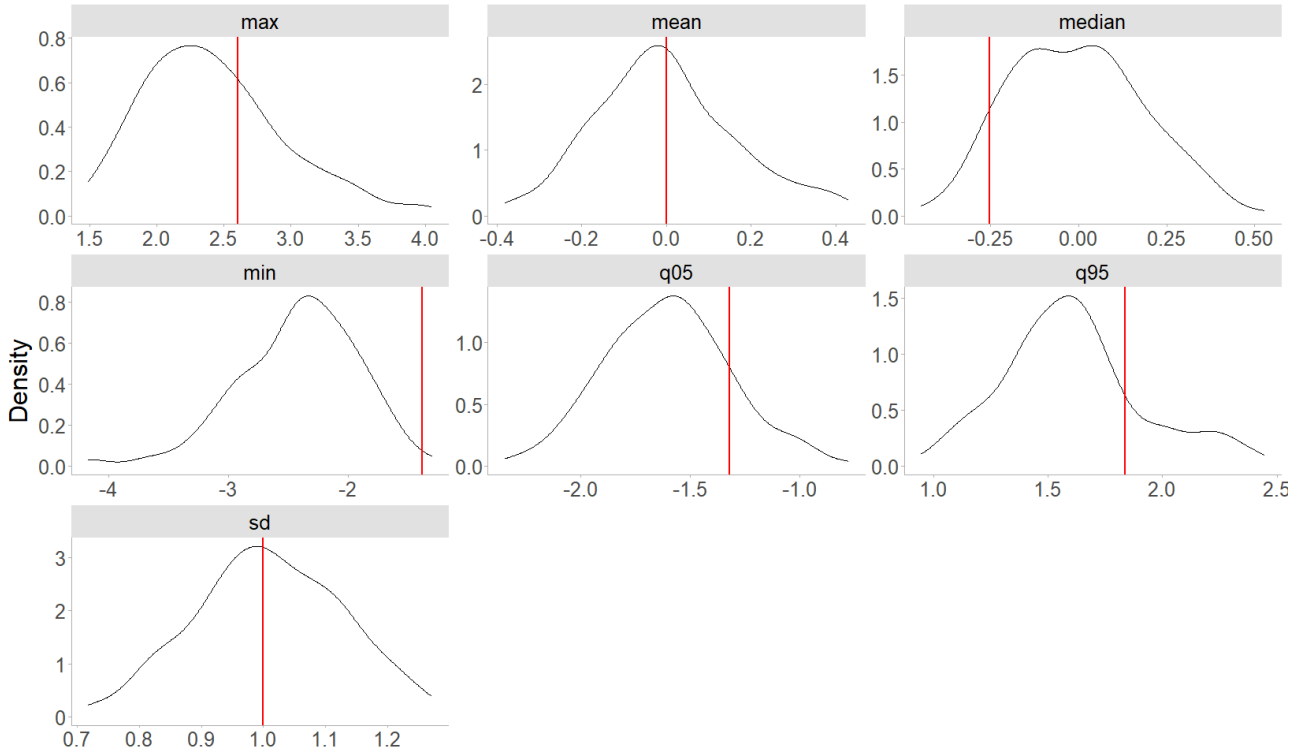



S_Figure 4: Posterior predictive check of the latent CTQ. The red line shows the observed values.

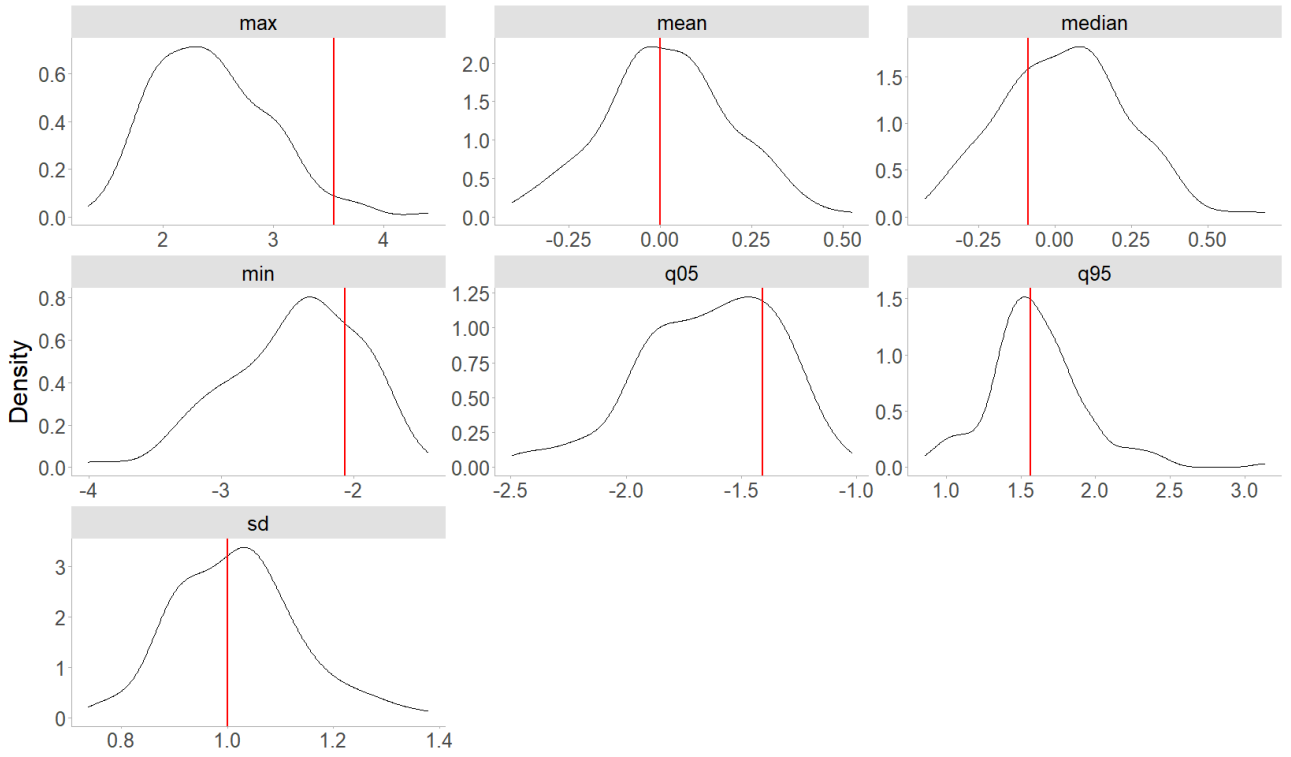
**Statistics**




S_Figure 5: Posterior predictive check of selected statistics of HCC. The red line shows the observed values.



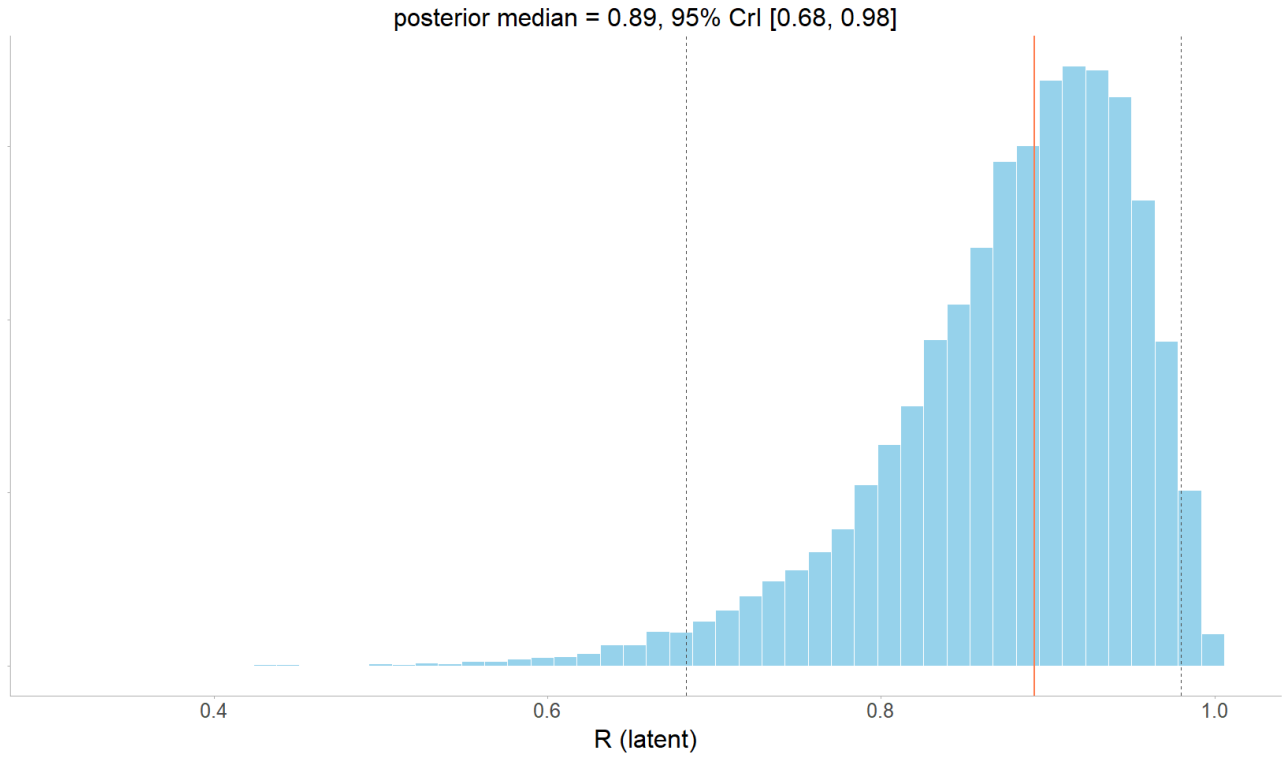

S_Figure 6: Posterior predictive check of selected statistics of PANSS. The red line shows the observed values.




S_Figure 7: Posterior predictive check of selected statistics of PSS. The red line shows the observed values.

**Latent trauma metric stratification**

CTQ subscales exhibit high mutual inter-correlation due to shared response variance. Modeling observed subscale scores as independent predictors introduces multicollinearity and leads to overestimating trauma burden when multiple subscales are elevated simultaneously (analogous to multiple motion sensors triggering from a single event). To resolve this, we employed a Bayesian implicit latent variable framework, modeling five observed CTQ subscale indicators ($\mathbf{y}_{\mathrm{EA}}, \mathbf{y}_{\mathrm{PA}}, \mathbf{y}_{\mathrm{SA}}, \mathbf{y}_{\mathrm{EN}}, \mathbf{y}_{\mathrm{PN}}$) as expressions of two correlated latent factors: Abuse ($c_1$) and Neglect ($c_2$).




S_Figure 8: Latent abuse–neglect correlation. The red line shows the posterior median and the dashed lines display the 95% credibility interval.

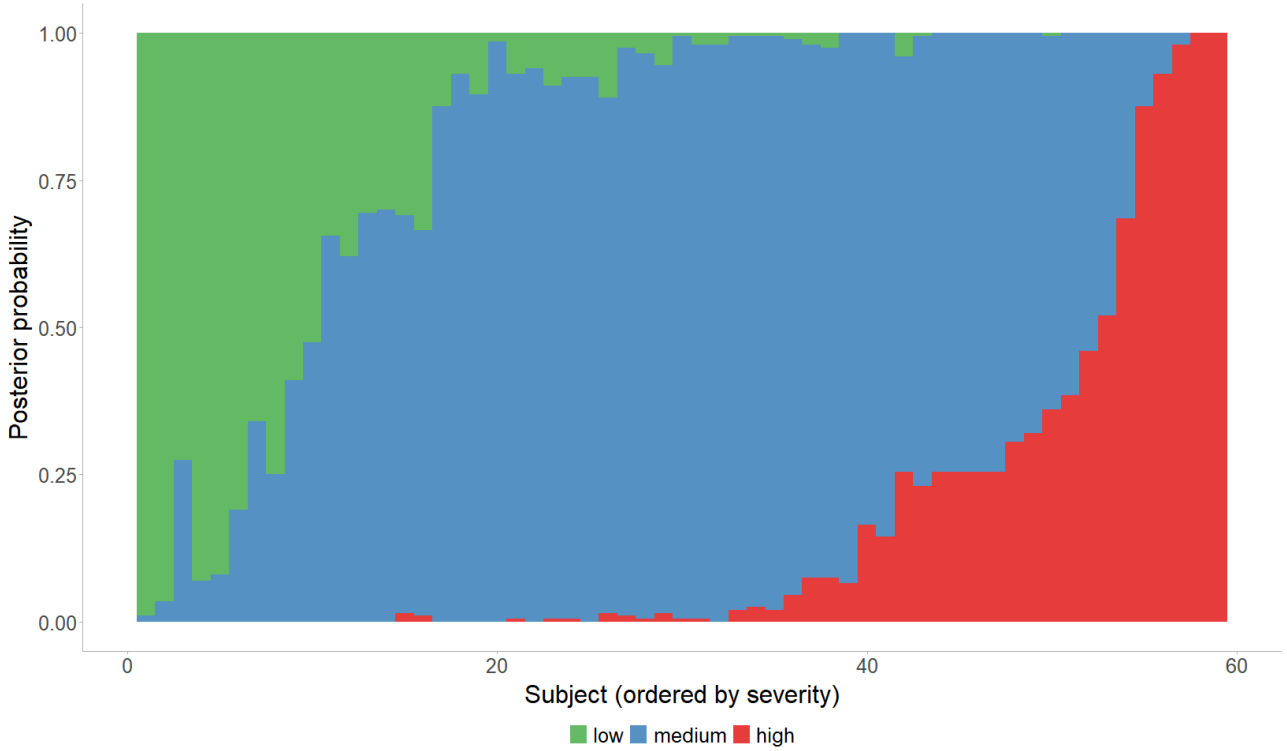
Mean latent factor score $(c_1, c_2)$ represents standardized, continuous latent trait category, by which the subjects were assigned into trauma severity tiers.

Low trauma: $\frac{1}{2}(c_1 + c_2) \leq -0.75$

Medium trauma: $-0.75 < \frac{1}{2}(c_1 + c_2) < 0.75$

High trauma: $\frac{1}{2}(c_1 + c_2) \geq 0.75$

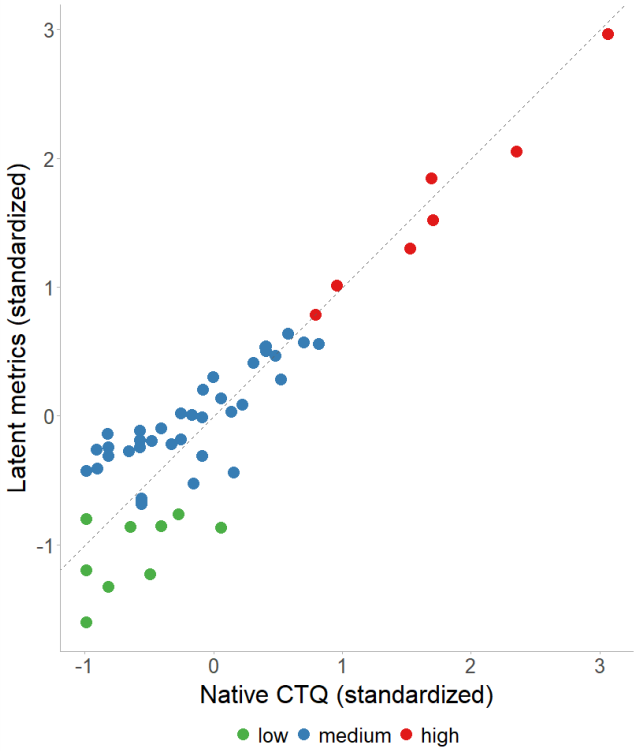
Because $c_{1,i}$ and $c_{2,i}$ are full posterior draws (not point estimates), each subject gets a posterior distribution over tiers rather than a single label.




S_Figure 9: Posterior trauma-tier posterior classification per subject - stacked posterior probability of low / medium / high, subjects ordered by mean latent severity.

Posterior predictive check of the latent metrics to the CTQ native scale could be displayed by converting the native CTQ using the following procedure:

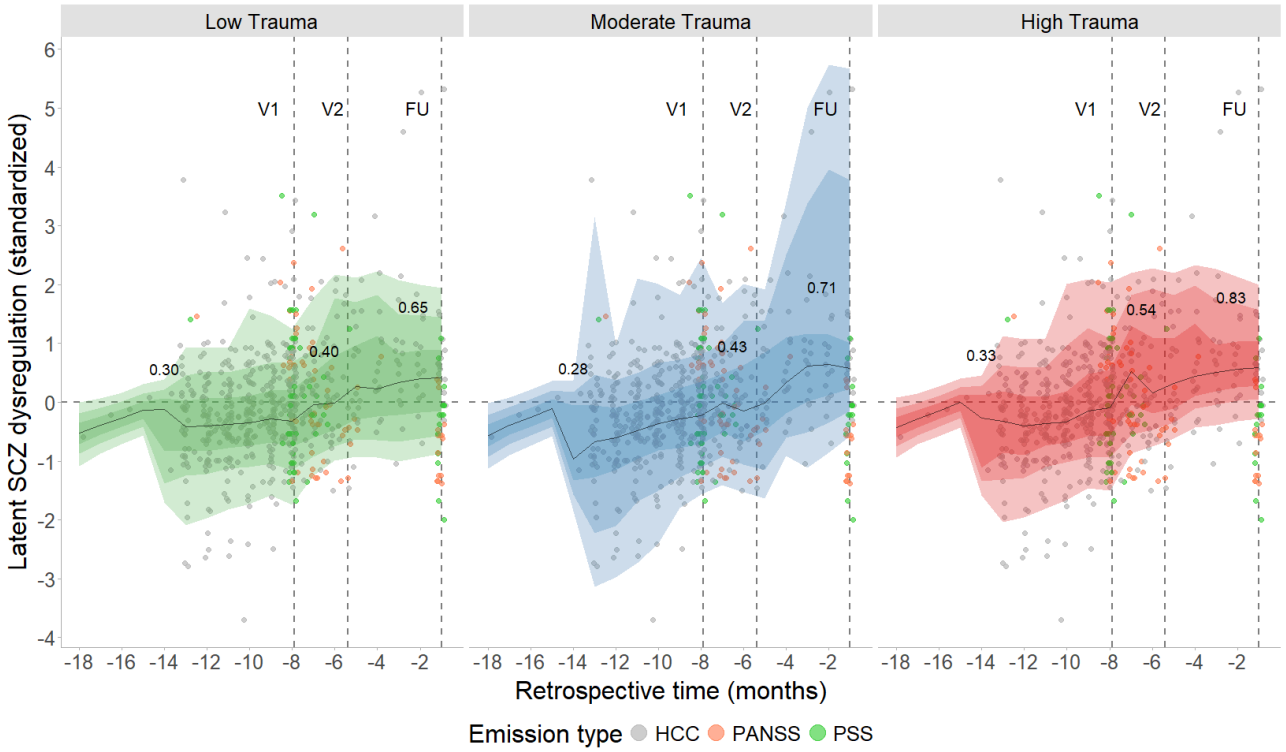
1. Taking a mean of the 3 abuse native subscales (EA, PA, SA).
2. Taking a mean of the 2 neglect native subscales (EN, PN).
3. Standardize each construct separately across subjects.
4. Averaging the two z-scores.




S_Figure 10: Native CTQ vs. posterior-mean latent metrics.

**Sensitivity analysis**

The original CTQ values were permuted across individuals and refitted into the model (S_Figure 11). The association with CTQ disappeared.




S_Figure 11: Retrospective trajectories stratified by childhood trauma severity across permuted subjects. Shaded areas represent 50%, 80%, and 95% posterior credible intervals.Numbers indicate the posterior probability that the latent state (z) is positive at the corresponding time points associated with each clinical visit, Pr(z(visit) > 0).